MDFF_NM Tutorial
Trajectory Analysis

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1.Introduction

After generating the structures using MDFF_NM, it is possible to project the structures: initial, final and generated in the space of the main Components to verify the sampling of the conformational space carried out by MDFF_NM. This analysis is carried out using a Pyhton script that uses the "Python Package for Protein Dynamics Analysis" developed by Prof. Dr. Ivet Bahar's group.T

The script developed in Pyhton reads the PDB referring to the initial structure used by the MDF_NM, the PDB referring to the target, a set of experimental PDB structures and the trajectories generated by running the MDFF_NM. As an output, it generates a graph with the projection of the generated and experimental structures in the space of the PCAs calculated from the experimental PDBs. An example is shown in Figure 1.

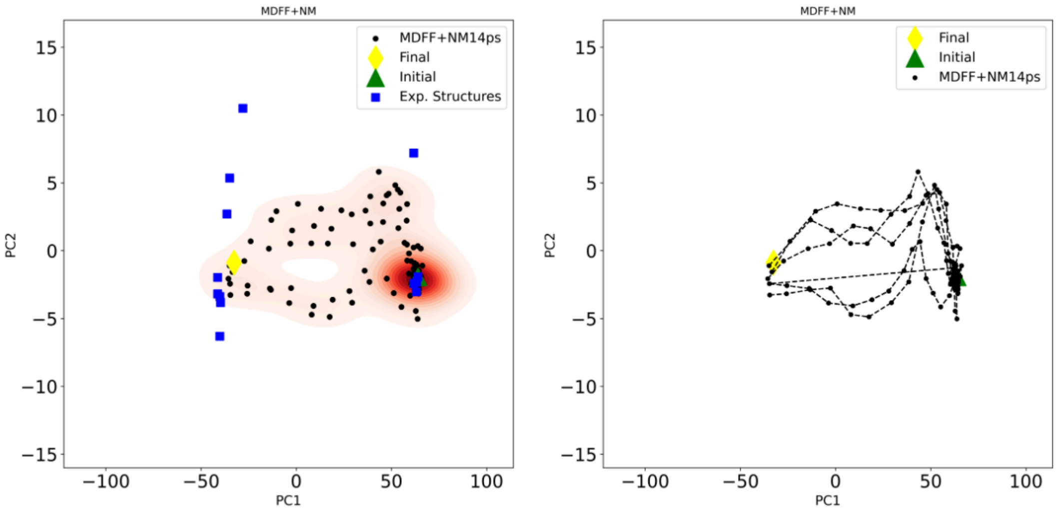

Figure 1- Example of the projections structures generated by MDFF_NM

2.Before starting

In the first step, you need to have installed on your computer :

- 1- **Pyhton 3.0**
- **2-Prody 2.4.1 : https://pypi.org/project/ProDy/**

Finally, you will need to use a development environment for Python, it can be google colab or jupyter-notebook or pycharmn.

Input Files:

- You need the dcds generated by the MDFF_NM
- You need the initial and final (Target) structure
- You need a set of experimental structures (PDBs) to compose a experimental DCD that will be used to calculate the PCAs.

3.Running Projection

Step 1- Prepare the inputs necessary

Step 2- open the script: projecao_mdfff-new-5traj

Step 3- Check the path and the files names of Input files in the code (Figure 2).

```python
#lendo dcd das trajetórias do MDFF com  Modos Normais

tmp_mdff1 = pr.parseDCD('./traj1.dcd')
tmp_mdff1.setAtoms(q_inicial)
tmp_mdff1.select('ca')
ens_mdff1 = pr.Ensemble(tmp_mdff1)
ens_mdff1.setAtoms(ens_exp.getAtoms())
ens_mdff1.setCoords(ens_exp)
ens_mdff1.addCoordset(tmp_mdff1.getCoordsets())
ens_mdff1.superpose()
mdff1_proj_2d = pr.calcProjection(ens_mdff1, pca_exp[:2], rmsd=False)

#lendo dcd das trajetórias do MDFF com  Modos Normais
tmp_mdff3 = pr.parseDCD('./traj2.dcd')
tmp_mdff3.setAtoms(q_inicial)
tmp_mdff3.select('ca')
ens_mdff3 = pr.Ensemble(tmp_mdff3)
ens_mdff3.setAtoms(ens_exp.getAtoms())
ens_mdff3.setCoords(ens_exp)
ens_mdff3.superpose()
mdff3_proj_2d = pr.calcProjection(ens_mdff3, pca_exp[:2], rmsd=False)
```

```
@> DCD file contains 27 coordinate sets for 214 atoms.
@> DCD file was parsed in 0.00 seconds.
```

```python
q_inicial = pr.parsePDB('./initial.pdb', subset='calpha', compressed=False)
q_final = pr.parsePDB('./final.pdb', subset='calpha', compressed=False)
tmp_exp= pr.parseDCD('./experimental.dcd')
tmp_exp
tmp_exp.setAtoms(q_inicial)
tmp_exp.select('ca')
```

```
@> 214 atoms and 1 coordinate set(s) were parsed in 0.01s.
@> 214 atoms and 1 coordinate set(s) were parsed in 0.01s.
@> DCD file contains 24 coordinate sets for 214 atoms.
@> DCD file was parsed in 0.00 seconds.
@> 0.06 MB parsed at input rate 69.49 MB/s.
@> 24 coordinate sets parsed at input rate 27518 frame/s.
```

Figure 2- reading the inputs

Step 4- Run new_scriptnew_script_projection using jupyter-notebbok or any python environment